CS556 Fall'25 Final Exam

Name:

Closed book: no textbook, no electronic devices, one sheet of paper with handwritten notes.

Read carefully before answering! Write your answers on the white sheets given to you. Show all your work, not only the final answer.

Question 1 *(10 points)* If possible, give a matrix A which has $\begin{bmatrix}1\\2\\1\end{bmatrix}$ as a basis for the columns space and $\begin{bmatrix}0\\3\\2\\-1\end{bmatrix}$ as a basis for its row space. If not possible, give your reason.

Question 2 *(10 points)* Let $A = \begin{bmatrix}1 & 4\\2 & 3\end{bmatrix}$. Find an invertible matrix S that makes $S^{-1}AS$ a diagonal matrix.

Question 3 *(10 points)* Each independent question refers to the matrix $A = \begin{bmatrix}4 & 1\\d & -4\end{bmatrix}$. In each case, find the value of d that makes the statement true (and show your work!).

(a) Give a value of d such that $\begin{bmatrix}5\\1\end{bmatrix}$ is an eigenvector of A.

(b) Give a value of d such that 2 is one of the eigenvalues of A.

(c) Give a value for d such that A is a nondiagonalizable matrix.

Question 4 *(10 points)* A rectangular storage container with an open top is to have a volume of $10m^3$. The length of its base is twice the width. The material for the base costs \$10 per square meter. Material for the sides costs \$6 per square meter. Find the cost of materials for the cheapest such container.

Question 5 *(10 points)* A social scientist decides to study in depth one individual to be randomly sampled from a given population. The social scientist is worried about the possibility that the chosen individual will refuse to cooperate. A previous student provides data on the probability of noncooperation for individuals in different income classes.

High Income	0.08
Medium Income	0.10
Low Income	0.04

The scientist knows that in the population the proportion of high income people is 0.25, of medium income people is 0.5, and of low income people is 0.25.

1. What is the probability that a randomly selected individual will be noncooperative?
2. Assuming that a randomly selected individual is noncooperative, what is the probability that that individual has low income?

Question 6 *(10 points)* If the function $f(x) = x^3 + ax^2 + bx$ has the local minimum value of $-\frac{2}{9}\sqrt{3}$ at $x = \frac{1}{\sqrt{3}}$, what are the values of a and b?

Question 7 *(10 points)* Consider the function: $f(x, y) = x^2y + 3sin(xy) + 2y^2$. We apply gradient descent with a fixed learning rate $\alpha = 0.1$ starting from the initial point $(x_0, y_0) = (1, 1)$. Using gradient descent, compute the points $(x_1, y_1), (x_2, y_2)$ and (x_3, y_3) reached after three iterations.

Question 8 *(5 points)* Find the derivative of $(4x^2 - 7)^{2+\sqrt{x^2-5}}$ using logarithmic differentiation.

Question 9 *(10 points)* The breakdown voltage of a randomly chosen diode of a certain type is known to be normally distributed with a mean value of 40 volt and standard deviation of 1.5 volt.

1. What is the probability that the voltage of a single diode is between 39 and 42?
2. What value is such that only 15% of all diodes have voltages exceeding that value?
3. If four diodes are independently selected, what is the probability that at least one has a voltage exceeding 42?

Question 10 *(15 points)* Find the values of a, b and c if the parabola $y = ax^2 + bx + c$, is to be tangent to the line $y = 4x + 1$, at the point $(-1, -3)$, and is to have a critical point at $x = -2$.

Question 11 *(0 points)* Would you like to be considered for a Course Assistant position for this course in Spring 2026?

- Yes
- No